

Blind Motion-Blur Parameter Estimation Using Edge Detectors

Robert Grou-Szabo

Department of Frontier Informatics, Graduate School of
Frontier Science, The University of Tokyo 7-3-1 Hongo,
Bunkyo-ku, Tokyo, 113-8656, Japan
grou@if.t.u-tokyo.ac.jp

Tadashi Shibata

Department of Electrical Engineering and Information
Systems, The University of Tokyo 7-3-1 Hongo, Bunkyo-ku,
Tokyo, 113-8656, Japan
shibata@ee.t.u-tokyo.ac.jp

Abstract — A method to determine the parameters of directional motion blurring noise using edge detectors is developed in this paper. Once the parameters characterizing the blurring noise are known, the point spread function (PSF) of the blurring noise can be reconstructed and the noise subsequently eliminated. The algorithm uses an edge detector with dynamic thresholding to extract edge information from an image at several threshold values. First the blurring angle is determined by rotating the image prior to edge extraction and the angle with the highest difference in edge count and its corresponding perpendicular angle is determined and considered to be the angle at which shifting has occurred. After the blurring shifting angle has been found, the pixel displacement length is determined using an iterative blind deconvolution process. Finally, once both the shifting angle and pixel displacement length are known the point spread function (PSF) of the blurring noise corrupting the image can be reconstituted.

I. INTRODUCTION

Digital photography has quickly become as commonplace in our daily lives as cell phones and PDAs. In most of the heavily populated urban areas, hardly a crime or public event can occur without someone taking a camera or cell phone out to take a picture of the event. Yet even with today's sophisticated cameras, it is not uncommon to take pictures that are too blurry to clearly distinguish anything or pictures where the subject is unintelligible due to poor lighting conditions, in which case the picture are unusable.

Perhaps the most prevalent type of noise to afflict amateur digital photography is blurring noise. There exist two kinds of blurring noise, anisotropic and isotropic blurring noise. Anisotropic blurring noise occurs when the camera is out of focus or the subject is outside of the camera's field of focus. Our research focuses on isotropic noise, also known as directional blurring noise, which occurs when the camera is shifted at the time of capture or when the camera is in motion. Two parameters are needed to characterize directional blurring noise corrupting an image they are; Shifting Angle and Pixel Displacement Length. The blurred image is modeled as a convolution between the original uncorrupted image and an unknown PSF modeling the blurring noise. Once the shifting

angle and pixel displacement are determined, the point spread function can be recreated and subsequently used for deblurring of the image. There are several algorithms that can be used to determine the parameters characterizing any blurring noise present in an image; however most of these techniques usually require complex mathematical tools and statistical analysis of the noisy image. Those methods can be effective in post production processing but are computationally intensive and therefore not suitable for real-time processing.

One of the initial suppositions of our research was that the algorithm must be "hardware-friendly" so as to be well suited to a VLSI implementation and furthermore integrated into a camera. The second suppositions was that shifting motion is linear and of uniform direction and velocity throughout the image. Therefore no rotation of the blurring parameters is expected, nor does the length at which blurring takes place vary throughout the image.

II. RELATED WORK

There exist several strategies that can be used to detect the presence of blurring noise in an image [1]–[6], recently the most of which are modified versions of either a Radon transform [7] or the Cepstral Method [8], however those methods require the use of computationally intensive mathematical tools such as Fourier Transforms or power spectrum analysis and so on. The work presented in this paper is geared towards real-time applications and more closely resembles the technique of "steerable filters" [9] to determine the parameters of blurring noise. The main difference is that in [9] steerable filters are swept over the power spectrum of the blurred image whereas our algorithm works directly with the image pixel intensities. In [10] Wang et al. use a technique that is similar to the one developed in this paper in order to determine an image quality measure. They do so by analyzing the gradient intensity at edges and by comparing sharp edges with those edges that have been smoothed out, they give an objective image quality metric. However they stop at giving a score to the image without giving the quantitative parameters of the noise corrupting the image.

In [13] an edge detector with a dynamic thresholding technique was used to determine the compression history of an image. The algorithm determines whether or not an image has ever been compressed, the technique is pushed farther and adapted in order to analyze the characteristics of blurring noise as well.

III. PROPOSED ALGORITHM

When a camera is shifted at the instant of capture, thereby causing directional motion blurring, all the edges that are parallel to the direction in which shifting has occurred will have a tendency to remain crisp and sharp. Whereas, edges that are perpendicular to the shifting angle will be smudged and affected the strongest, resulting in fuzzy and blurred edges. The edges in an angle perpendicular to shifting will therefore have the least amount of distinct and sharp edges. Therefore, by searching for the angle that has the greatest disparity in the amount of edge information between an angle and its corresponding perpendicular angle, the angle at which shifting has occurred at the time of taking the picture can be determined. After the angle has been determined the blurring pixel displacement length is estimated by an iterative deconvolution process. Once both the angle and displacement length have been determined the PSF can be reconstructed to subsequently denoise the image with better results.

A. Shifting Angle Determination

Fig. 1-A is an example of the effects of blurring noise on an image and Fig. 1-B is the edge map of the same image filtered with a Prewitt horizontal edge detector. Although strictly speaking only the horizontal edges are exposed, we can see several line segments in 1-B appear to be at an angle and parallel to each other. The lines are actually dithered

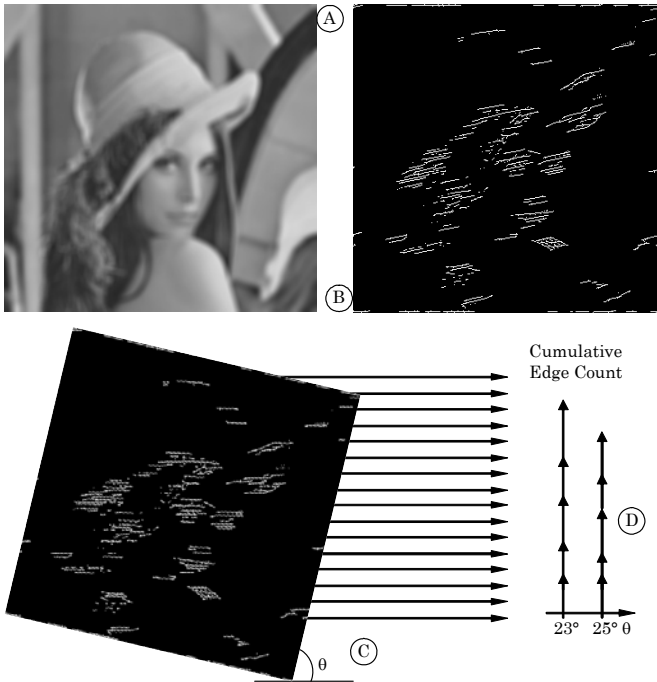


Figure 1. A) Effect of directional blurring noise. B) Horizontal edge detector engenders dithered slanted edge features. C) Rotated edge map. D) Cumulative edge count at each angle of rotation

horizontal lines and the slanted segments that the steps portray follow the same direction in which shifting has occurred.

By rotating the image prior to edge detection as in Fig. 1-C, we can determine the angle that results in the greatest disparity in the amount of cumulative edge information between an angle and its perpendicular angle. That angle is the angle at which shifting has occurred. In order to use all the edge information available hidden in the image we apply the edge detector at several threshold values and consequently obtain several estimates for the blurring angle.

Figure 2 illustrates how the data is accumulated and processed. In the top graph of figure 2 the horizontal axis is the angle of rotation before applying an edge detector to an image. The vertical axis is the cumulative edge count at an angle (as seen in Fig. 1-D) where each stratified line of the graph represents the edge detector applied at a different threshold value. Fig. 2-E is an angle chosen at random and Fig. 2-F is the perpendicular angle. Fig. 2-G is the differential in edge count of an angle with its corresponding perpendicular angle. The peak at Fig. 2-G is the point where there is the largest disparity between the edge count at an angle and the edge count at its perpendicular. It is at this point G that points to the angle that shifting has occurred.

In our experiments we used the median value of all the estimated angles as the final estimate; however there are other possible methods for estimating a representative value chosen from among a sample set. Something that is rarely discussed in papers is the certainty of an algorithm's estimation process. The spread of all the peaks provides a confidence range, for

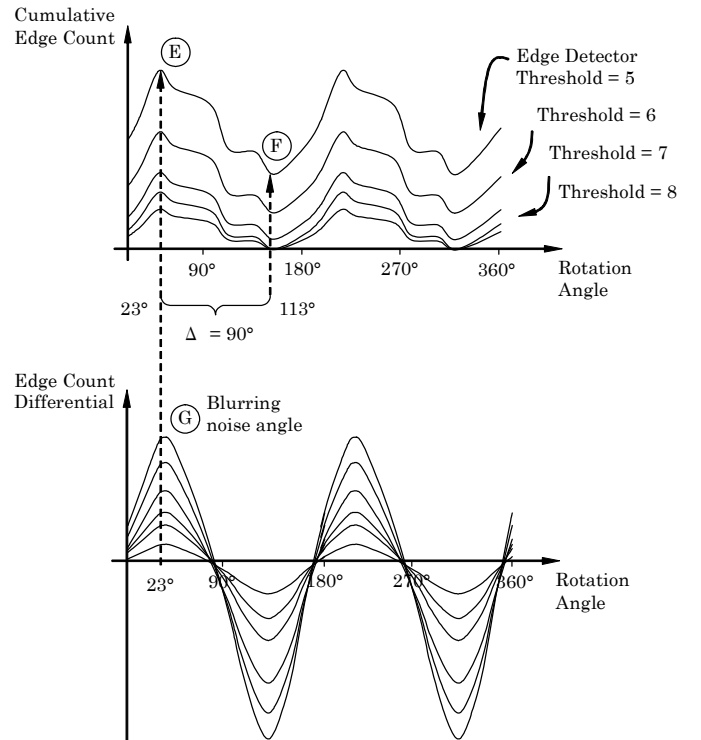


Figure 2. Top: Horizontal axis, image rotation angle. Vertical axis, raw cumulative edge data of an edge detector. Each stratified line is the edge detector applied at a different threshold value; Bottom: Differential edge count of an angle with its corresponding perpendicular angle

example if all the peaks at Fig. 2-G for every threshold value all line up and are within one standard deviation of each other then we can be more confident of the final estimate, whereas if the disparity of the estimates is wide, in other words the peaks at Fig. 2-G are flat, wide and sparse, it means that the blurring noise was not very strong and there is very little difference in the edge count between an angle and its perpendicular angle. This informs us that the blurring pixel displacement is not very large. As the intensity of blurring noise increases, so do all the peaks at Fig. 2-G line up and become narrower pointing towards the same angle. The first step in determining the blurring noise parameters is to find the blurring angle. The pseudo code describing this part of the algorithm is below:

```

Load TestImage;
For (EdgeThreshold = 0 to 255) ← Step A1
{
  For (ImageAngle = 0 to 360) ← Step A2
  {
    RotatedImage = RotateImage(TestImage, ImageAngle); ← Step A3

    Horiz_EdgeMap = Prewitt Edge (RotatedImage, Thresh, horizontal); ← Step A4
    Vert_EdgeMap = Prewitt Edge (RotatedImage, Thresh, vertical); ← Step A5

    HorizEdgeCount = flatten(Horiz_EdgeMap); ← Step A6
    VertEdgeCount = flatten(Vert_EdgeMap); ← Step A7

    ΔThreshold(ImageAngle) = Abs(HorizEdgeCount - VertEdgeCount); ← Step A8
  }
  AngleGuessAtThreshold(EdgeThreshold) = Greatest(ΔThreshold); ← Step A9
}
BlurringAngle = mean value(AngleGuessAtThreshold); ← Step A10

```

Pseudo code

- Step A-1) The algorithm is applied to an image using the complete range of threshold values of an edge detector, spanning from 0 to 255. This is done so that every possible edge in the image is taken into account. It is worth noting that when the edge detector is applied to an image that is at an angle, a contour edge appears all around the boundary of the image. This boundary edge interferes with the flattening of the edge map in Step 6 & 7 and therefore a 1 pixel strip around the rotated image must be removed before flattening the edge map.
- Step A-2 & A-3) The image is rotated around its center for 360 degrees. The precision of the rotation angle also determines the step size and refinement with which the angle can be estimated. [See Fig. 2-G]
- Step A-4 & A-5) A horizontal edge detector is passed over the image at every possible threshold value (as described in Step 1). A single horizontal edge detector is necessary if the image is rotated 360 degrees, however the image need only be rotated 90 degrees if 2 perpendicular edge detectors are used at the same time since the second half of the cumulative edge data is simply a repetition of the first half. Therefore, if we want to minimize the use of rotation matrices, both a horizontal and vertical edge detector can be used and the image rotated by 90 degrees. Then by repetition the edge map for the complete 360

degrees can be collated by appending the result of the first 90 degrees from the horizontal edge map to the result of the vertical edge map of the first 90 degrees, then the resulting waveform duplicated and appended to itself. [See Fig. 2-Top] In this way, we can obtain the cumulative edge information at one threshold value of all 4 quadrants of an edge map simulating a rotation of the image by 360 degrees. In our simulations the Prewitt directional edge detector gave the best results.

- Step A-6 & A-7) At every edge detector threshold value and for every rotating angle, the edge map is flattened and the cumulative edge count is summed. Essentially calculating how much "edge information" is contained in an image for that rotation angle and edge detecting threshold value. [See Fig. 1-D]
- Step A-8) Next we determine the angle with the highest edge count for that threshold value. The angle with the largest difference in edge count between itself and its perpendicular direction is the direction in which shifting has occurred. [See Fig. 2-E & F]
- Step A-9) In Step A-8 an estimate of the angle is determined and Step 9 groups all the angular estimates into an ordered array. Each slot of the array represents the estimate for that particular threshold; therefore we obtain 256 estimates for the angle, estimates with zero edge information are not counted. [See stratified lines of Fig. 2]
- Step A-10) The final step simply consists of going through the array of Step 9 and finding the angle that is repeated the most often, or the median value can also be calculated. This will be the angle at which blurring motion occurred. [See Fig. 2-G]

B. Pixel Displacement Determination

After the shifting angle has been determined, next the pixel displacement length is estimated. The method for doing this, called blind deconvolution, has been well studied. The Richardson-Lucy algorithm is a well-known image restoration technique for example used in astronomy to clear up blurry images [11]; however even in that case a human is still needed to determine if the resulting image, after deconvolution, is subjectively "nicer" than before. However we wish to develop an algorithm that is completely autonomous. The aim is to remove the human from the loop and produce an algorithm that requires no human intervention whatsoever.

In order to automate the process of determining the best estimate of the pixel displacement is to first start the process with an initial estimate of 3 pixels in the direction of the angle previously determined. Then using the Richardson-Lucy algorithm the image is deconvoluted and the resulting image is analyzed. After which an edge detector is swept over the convoluted image where periodic wave-patterns can be observed. When these periodic patterns are present, it means that the parameters used in the deconvolution process were not identical to original parameters characterizing the blurring noise corrupting the image. However when the parameters

used in the R-L. algorithm are identical to the original blurring noise parameters, those wavy patterns are almost non-existent in the deconvoluted image. When the convolution is not well conditioned periodic wavy lines appear. These are what need to be minimized in order to determine the pixel displacement value. An image with many wavy artifacts is the result of an incorrect estimated input argument of the pixel displacement and translates into an edge map, after deconvolution of the image, with a lot of edge information. Conversely a proper estimate of the noise parameters results in a lower edge count on the edge map after deconvoluting the image.

An initial estimate for the pixel displacement of 3 pixels is used because any blurring noise with a pixel displacement of less than 3 pixels would hardly be noticeable and any subsequent deblurring would corrupt the image more than clearing it up. Therefore an initial estimate of 3 pixels for the blurring displacement is used as the lower boundary of detectable blurring noise.

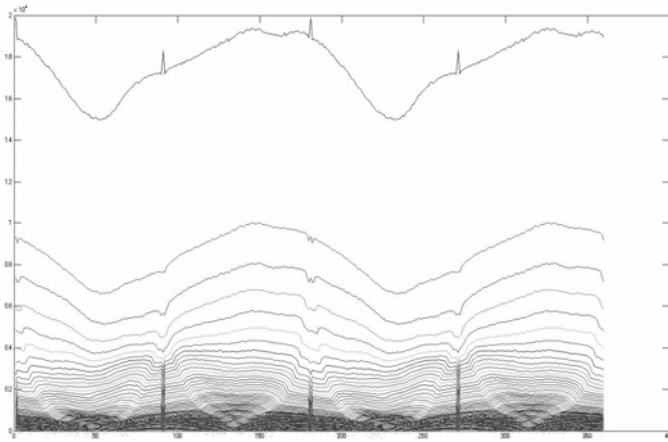


Figure 3. Raw data from the Shifting Angle Determination phase; Horizontal axis is the angle of rotation, Vertical is the cumulative edge count of a prewitt edge detector. The stratified lines are from the incrementing threshold of the edge detector

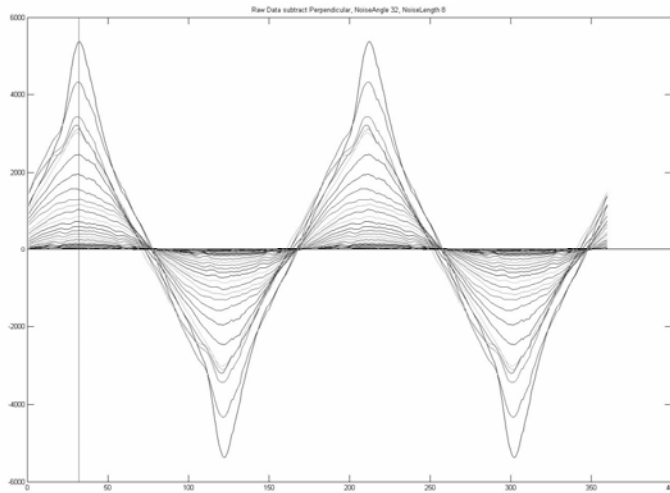


Figure 4. Horizontal axis is the angle of rotation, Vertical is the differential cumulative edge count between an angle and its corresponding perpendicular angle. The peak identifies the blurring noise shifting angle.

By iteratively increasing and decreasing the input parameter of the pixel displacement value and reapplying the R-L deconvolution algorithm, the amount of edge data varies according to how close or how far the estimate is to the real value. The local minimum value on this cumulative edge count is sought. At each iteration after the execution of the Richardson-Lucy algorithm, and an edge detector passed over the resulting image, then the edge information is cumulated and indexed according to the displacement value used as the input parameter for the Richardson-Lucy algorithm. It can iteratively be done in this way because the shifting pixel displacement is an integer and therefore there are only a limited number of possibilities, which can be verified exhaustively starting from the lower limit boundary of ± 3 pixels. When the minimum in the edge count is determined, then the value of the input argument used for the pixel displacement for that iteration cycle is considered to be the value for the pixel displacement of the blurring noise corrupting the image.

IV. RESULTS

Matlab simulations were executed using images depicting several different types of scenery which were corrupted with blurring noise at random angles and/or pixel displacement lengths. Using this algorithm we have achieved an 84% success rate for correctly identifying the shifting angle for various images at degrees of blurring noise, however the success rate for automatically determining the pixel length is slightly higher and stands at about 91%. This is due to the fact that the precision of the pixel displacement is an integer value, i.e. 4, 8 or 10 pixels of displacement. Whereas the shifting angle can take on any real number, i.e. 37.61 degrees.

Figure 3 shows the raw data from blurring noise shifting angle determination phase. Each stratified line is the result obtained by rotating an edge detector at each threshold value producing 255 estimates of the shifting angle resulting in greater accuracy; among these we take the median value as the final result. Figure 4 is the graph of the differential edge count between an angle and its perpendicular angle. The vertical line that crosses the graph at each peak is the estimated shifting angle of the blurring noise.

Figure 5 shows the result obtained by iteratively applying the Richardson-Lucy algorithm with an increasing pixel displacement value (horizontal axis) and then pass an edge detector over the resulting deconvoluted image and draw the graph of the cumulative edge count at each incremental for the input parameter of the R-L. algorithm. The local minimum dip is where there is the least amount of edge information and also the point at which there are the fewest periodic wavy patterns in the deconvoluted image. This is the closest value for the pixel displacement of the blurring noise corrupting the image. Figure 5-A and 5-B are results taken from an image that has been corrupted by blurring noise with different displacement lengths.

V. CONCLUSION

Some of the intrinsic properties of the image being analyzed can also affect the success of determining the parameters of the blurring noise. For instance the amount of edge information contained in the image will affect the result of the algorithm. An image depicting a sky or calm water has little edge information and therefore even when blurry will still be pleasing to the eye. Conversely an image that has a lot of edge information and complicated patterns and lines will be easy to notice any blurring noise will be much easier to detect.

The second property that can affect the efficiency of the algorithm is the image size. In the case of a small image such as a desktop icon, each pixel is very different from its surrounding neighbors and as such edges are more difficult to detect. In this case also the algorithm does not do so well, however any blurring noise of a small image will quickly render it undecipherable.

The third parameter that reduces the effectiveness of the algorithm is the pixel displacement length. Below a certain threshold there is little difference in edge count between an angle and its perpendicular angle. However if the blurring happens to be below this threshold, then the impact on the images is also almost non-existent and perhaps even unnoticeable.

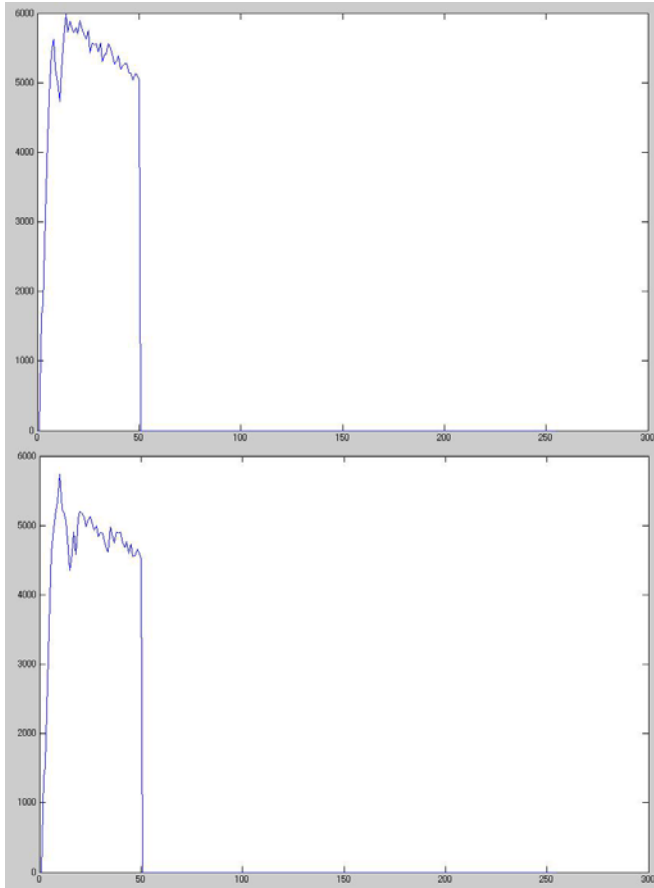


Figure 5. Blurring noise pixel displacement phase. Top: Cumulative edge count of an edge detector after applying the Richardson-Lucy algorithm; Bottom: Same as top with a different blurring noise pixel displacement.

An interesting fact about the nature of our world is that in most cases there will usually be a higher edge count in an image both horizontally and vertically. This is due to the horizon, roads, a flat body of water or else vertical skyscrapers, trees and telephone poles etc. Therefore there are always spikes in edge count both at 0 and 90 degrees when rotating an edge map, however this spike will be [overlooked, eliminated] ignored since we are searching for the angle with the greatest difference in edge count, therefore the spikes in edge count at 0 degrees and 90 degrees will cancel each other out.

Addressing computational complexity: The two operations of this algorithm that are computationally complex are; rotating the image (using rotation matrices) and developing the Lucy-Richardson deconvolution algorithm. The first, rotation of the image, need not be as complicated as one might think. For ease of explanation the image is rotated however the edge detector kernel can also be rotated in the opposite direction instead achieving the same result. As for the hardware implementation of the Richardson-Lucy algorithm, an effective FPGA implementation was developed in [12] using parallel hardware acceleration techniques.

In Step A-1 of the Shifting angle determination phase the algorithm is described as going from 0 to 255 however there is very little edge information at thresholds greater than 100. Therefore in order to accelerate computation it is recommended to stop the edge detector at thresholds greater than 100.

Also, the angle of rotation in Steps A-2 & A-3 range from 0 to 360 but can also be done with jumps of more than 1 degree. Because the pixel displacement is an integer, there are many angle values that do not need to be repeated. In Figure 6 the same operations as were done in Figure 4 however in this case it was done in two steps, first a coarse step using jumps of 10 degrees to roughly locate the angle and then a second finer step to more accurately pinpoint the proper blurring noise shifting angle. This reduces the number of operations by a factor of 6.

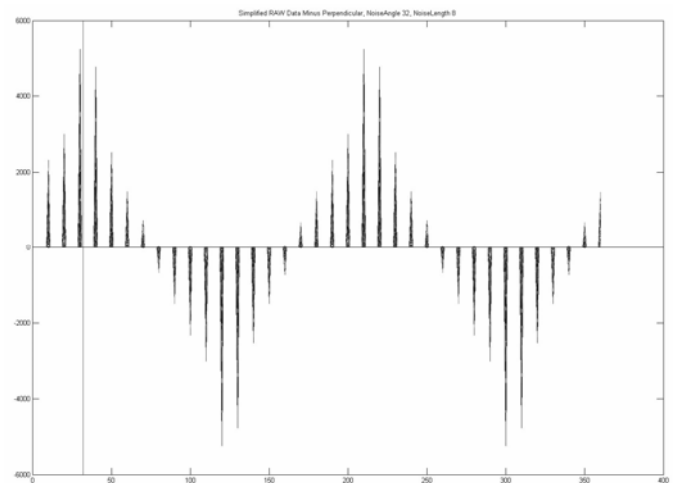


Figure 6. Identical to Figure 4 with angles taken at every 10 degrees

REFERENCES

- [1] V. Kayargadde and J. B. Martens, "Estimation of perceived image blur using edge features", *Int. J. Imag. System Tech.* 7 (1996), pp. 102–109
- [2] J. P. Oliveira, Mario A. T. Figueiredo, J. M. Bioucas-Dias, "Blind Estimation of Motion Blur Parameters for Image Deconvolution", in *Pattern Recognition and Image Analysis, LCNS Vol 4478/2007*, Springer-Verlag, Berlin Heidelberg, pp. 604 – 611, July 2007.
- [3] Hui Ji, Chaoqiang Liu, "Motion blur identification from image gradients", in: *Computer Vision and Pattern Recognition, 2008. IEEE Conference on, (CVPR 2008.)*, pp. 1-8, June 2008.
- [4] Li, Q., Yoshida, Y., Parameter-Estimation and Restoration for Motion Blurred Images, *IEICE(E80-A)*, No. 8, August 1997, pp. 1430-1437
- [5] Yitzhaky, Y., Kopeika, N.S., Identification of Blur Parameters from Motion Blurred Images, *GMIP(59)*, No. 5, pp. 310-320, September 1997
- [6] Kundur, D. Hatzinakos, D., "Blind Image Deconvolution", *Signal Processing Magazine, IEEE*, Volume 13, Issue 3, May 1996 Page(s):43 - 64
- [7] M. E. Moghaddam and M. Jamzad, "Finding point spread function of motion blur using radon transform and modeling the motion length," in *Proc. 4th IEEE International Symposium on Signal Processing and Information Technology (ISSPIT '04)*, pp. 314–317, December 2004
- [8] Shiqian Wu, Zhongkang Lu, Ee Ping Ong, Weisi Lin, "Blind Image Blur Identification in Cepstrum Domain", in *Proc. of 16th International Conference on Computer Communications and Networks, 2007. (ICCCN 2007)*, pp. 1166-1171, Aug. 2007
- [9] Ioannis M. Rekleitis, "Steerable filters and cepstral analysis for optical flow calculation from a single blurred image", *Vision Interface*, pp. 159-166, May 1996
- [10] Xin Wang, Baofeng Tian, Chao Liang, Dongcheng Shi, "Bling Image Quality Assessment for Measuring Image Blur", *Image and Signal Processing, 2008. CISP '08. Congress on*, May 2008, pp. 467-470
- [11] L. B. Lucy, "An iterative techniques for the rectification of ob-served distributions," *Astronomical Journal*, vol. 79, no. 6, pp.745–754, 1974
- [12] Sims O., Irvine J., "A real-time implementation of Richardson0Lucy deconvolution", *International Symposium on Field Programmable Gate Arrays, Proc. ACM/SIGDA*, 2006
- [13] R. Grou-Szabo and T. Shibata, "Blind image compression history determination using dynamic thresholding," in *Proc. ICASSP 2008*, pp. 1005-1008, April 2008